Is there hysteresis in potential output estimates?*

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Abstract

Using an error correction model, this paper studies the dynamics of GDP and potential GDP using OECD's Economic Outlook and IMF's World Economic Outlook estimates of the output gap. The findings suggest the presence of hysteresis, implying that short-run shocks to GDP have sizeable permanent effects on potential GDP.

Keywords - Hysteresis, Business cycle, Potential GDP

JEL Code - E32, E37, E47, E58

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1. Introduction

The *natural rate hypothesis* posited by Friedman (1968) is composed of two statements. First, monetary policy cannot affect permanently the unemployment rate or GDP, and second, if it tries to do so, it will face higher inflation. O. Blanchard (2018) provided a toy model to understand what the first statement means. He suggested that GDP (in logs, y_t) can be decomposed into potential GDP (y_t^*) and the output gap (y_t) and that $y_t^* = ay_{t-1}^* + \eta y_{t-1}$. The natural rate hypothesis holds when $\eta = 0$. This hypothesis, once controversial, has become a central assumption in economics. However, recent economic crises cast doubts on this assumption and there is evidence and theory suggesting that $\eta > 0$ (see Cerra et al. (2023)).

This letter contributes to the debate on the size of η by estimating it for two commonly used real-time estimates of y_t^* found in the IMF's World Economic Outlooks (IMF's WEO) and the OECD's Economic Outlooks (OECD's EO). To do so an error correction model, similar to the one used in the Ball & Onken (2022) study on NAIRU hysteresis, was estimated on y_t and these two measures of y_t^*. The main results suggest that there is hysteresis in these two commonly used estimates of y_t^*. For instance, a 1-percentage point demand shock is associated, on average, with an increase in potential GDP of about 0.93 percentage points with a 95% CI from 0.41 to 1.45 in the long-run. These findings suggest that expansionary fiscal policies could be self-financing (see Delong et al (2012)) and imply that monetary policy may not be neutral in the long run (see Jordà et al. (2020)).

2. Methodology

It is known that y_t and y_t^* behave as a unit-root process. Moreover, in equilibrium, $y_t = y_t^*$. Thus, these variables are cointegrated. The following model, inspired by Ball & Onken (2022), is proposed to study hysteresis in y_t^* real-time estimates.

$$\Delta y_{c,t} = \gamma_0 \Delta y_{c,t}^* + \gamma_1 \Delta y_{c,t-1}^* + \gamma_2 \Delta y_{c,t-1} + \gamma_3 \left(y_{c,t-1} - y_{c,t-1}^*\right) + \alpha_c + \tau_t + \upsilon_{c,t} \qquad (1)$$

$$\Delta y_{c,t}^* = \beta_1 \Delta y_{c,t-1}^* + \beta_2 \Delta y_{c,t-1} + \beta_3 \left(y_{c,t-1} - y_{c,t-1}^*\right) + \mu_c + \delta_t + \varepsilon_{c,t} \qquad (2)$$

Where Δ represents the first difference. μ_c and α_c are country-fixed effects. δ_t and τ_t are time-fixed effects. If there is a tendency towards equilibrium, $\gamma_3 < 0$. If there is hysteresis,

$\beta_3 > 0$. The inclusion of $\Delta y^*_{c,t}$ on equation 1 allows $\boldsymbol{y^*_{c,t}}$ to have a contemporaneous impact on $y_{c,t}$. $\upsilon_{c,t}$ is a demand shock, and $\varepsilon_{c,t}$ is a supply shock.

The degree of hysteresis is measured as follows (see Jordà et al. (2020)):

$$\eta = \frac{\text{Long} - \text{run effect on } y^*_{c,t} \quad 20\text{th year}}{\text{Cumulative sum of the output gap}} \tag{3}$$

from a 1-unit demand shock ($\upsilon_{c,t}$). The system (1-2) was estimated using seemingly unrelated regressions. Monte Carlo simulations using the covariance matrix were performed to extract the standard deviation of η and the long-run effect on $y^*_{c,t}$.

3. Data

The study used data on IMF's WEO and OECD's EO output gap estimates. Table 1 shows the period and country coverage of each source.

Table 1: Data sources

Sources	Year coverage	Countries (ISO Codes)
IMF's WEO	2003-2022	AUS, AUT, BEL, CAN, FIN, FRA, DEU, IRL, ITA, JPN, NLD, NZL, NOR, PRT, ESP, SWE, GBR, and USA.
	2009-2022	CYP, DNK, GRC, MLT, SVK, SVN.
	2012-2022	EST, KOR and LUX.
OECD's EO	1996-2021	AUS, AUT, BEL, CAN, DEN, FIN, FRA, DEU, GRC, IRL, ITA, JPN, NLD, NOR, PRT, ESP, SWE, CHE, GBR, and USA.
	1997-2021	NZL.
	2000-2021	ISL.

Note: The data used to construct real-time estimates of IMF's WEO used the September or October version from 2003 to 2021, OECD's Economic Outlook, Fall versions from 1996 to 2022 versions.

Transformations were made to get $y^*_{c,t}$ time series. In the case of output gap estimates $(\widehat{Y_t})$ they are measured as the percent deviation from $Y^*_{\boldsymbol{t}}$. Therefore, $Y^*_{\boldsymbol{t}}$ (in levels) can be recovered by the following formula:

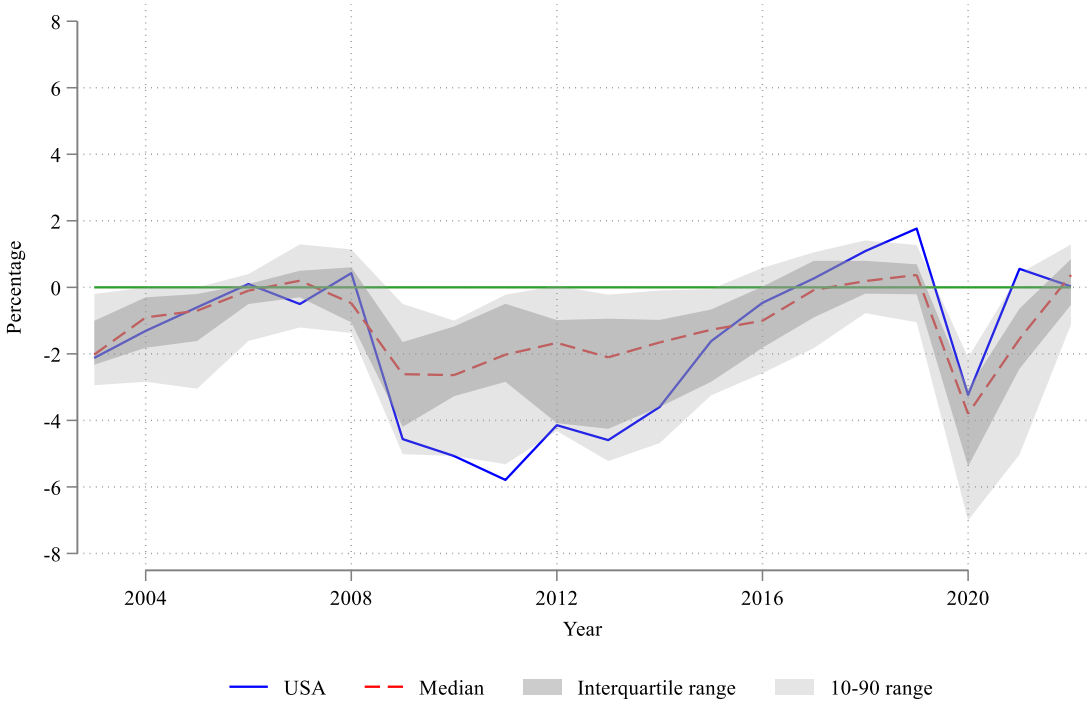
$$Y^*_{\boldsymbol{t}} = \frac{Y_t}{1 + \widehat{Y_t}/100} \tag{4}$$

where Y_t is real GDP in national currency. To assure consistency, real GDP data was taken from the 2022 version of each source.

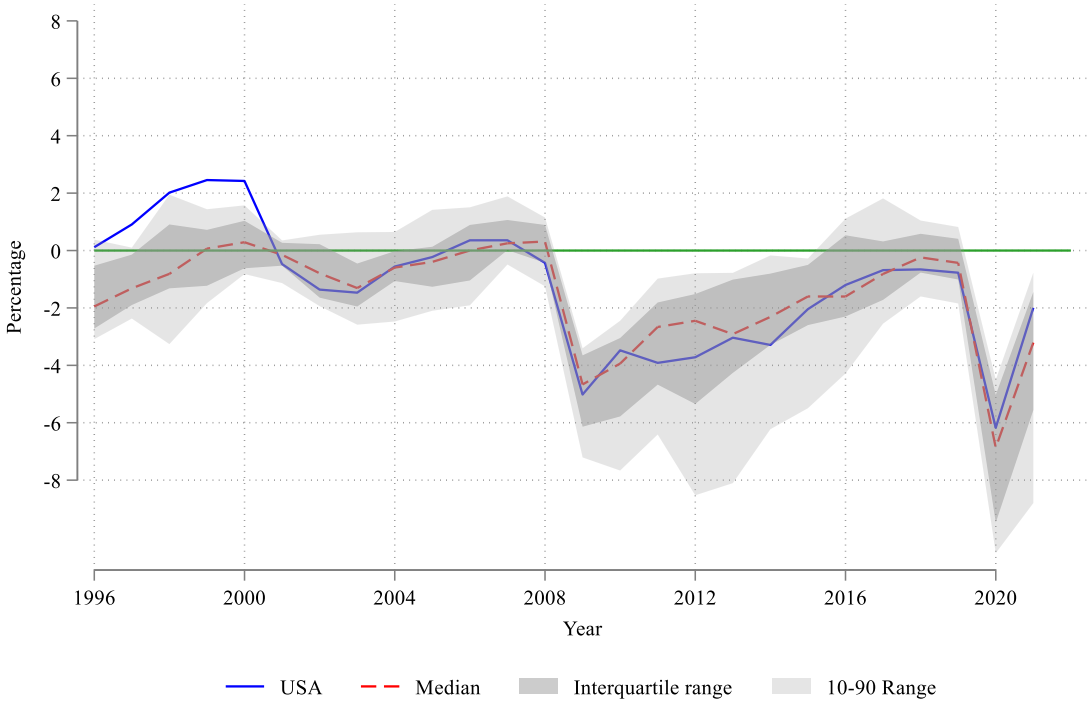
Figure 1 shows the estimates of the output gap estimates across sources. Panel (a) shows the dispersion of the IMF's WEO real-time output gap estimates across time. From 2008 to 2015, most output gap estimates were negative indicating slack in these economies. From 2016 to 2019, the economies were close to full employment until the COVID-19 crisis hit and depressed the economies. A similar story is told by the OECD's EO real time estimates (see panel b), the main difference is the severity of the global financial crisis and the COVID-19 crisis. The OECD's EO data also covers the .com crisis.

Figure 1: Output gap estimates across sources

(a) IMF's WEO real time output gap estimates



(b) OECD's EO real time output gap estimates



Source: Own calculations

4. Results

Table 2 shows the main results. As expected, $\gamma_3 < 0$ at a 1% significance level, which points to a tendency of GDP (y_t) to wander around potential GDP (y_t^*). Moreover, $\beta_3 > 0$ at a 1% significance level, which suggest the presence of hysteresis, that is, potential GDP is affected by demand shocks.

On average, the long-run effect of a unit demand shock is sizeable as it is close to 0.94. The long-run effect is higher when using OECD's figures and could be higher than the unity. The long-run effect found with IMF's figures is around 2/3 of the initial shock, which is still a large effect.

Regarding the hysteresis elasticity, the estimates are close between measurements. In comparison to previous literature, the estimates are the in medium range. Delong et al (2012) argued that the hysteresis elasticity should lie around the 0-0.25 in the United States. Rawdanowicz et al. (2014) found that η could range from zero to 1.2 on OECD

economies. Finally, Jordà et al. (2020) found that η was about 0.67 on developed economies.

Table 2: Main estimates

	IMF's WEO	OECD's EO
Equation 1		
γ_0	0.862***	0.872***
	(0.0275)	(0.0358)
γ_1	0.0354	0.0472
	(0.0469)	(0.0466)
γ_2	0.151***	0.260***
	(0.0467)	(0.0450)
γ_3	-0.524***	-0.288***
	(0.0441)	(0.0312)
Observations	423	523
R-squared	0.907	0.855
Equation 2		
β_1	-0.0618	-0.164***
	(0.0829)	(0.0565)
β_2	0.149*	0.121**
	(0.0822)	(0.0548)
β_3	0.309***	0.295***
	(0.0765)	(0.0358)
Observations	423	523
R-squared	0.456	0.538
η	0.339***	0.2826***
	(0.0848)	(0.0318)
Long-run effect	0.6748***	1.2103***
	(0.1860)	(0.2190)

Source: Own calculations using data from OECD's Economic Outlooks, IMF's World Economic Outlooks.
Note: Standard errors in parentheses, *** p < 0.01, ** p < 0.05, * p < 0.1, OECD's EO columns represent the estimation done with OECD's real-time output gap data, IMF's WEO columns represent the estimation done with IMF's World Economic Outlook real-time output gap data. All regressions include country fixed-effects and year fixed-effects.

5. Conclusions

This paper asked if there was hysteresis in $y^*_{c,t}$ real-time estimates. To do so, an error correction model was applied to two commonly used $y^*_{c,t}$ real-time estimates, following Ball & Onken (2022) study. The findings suggest that there is hysteresis in both IMF's and OECD's real time $y^*_{c,t}$ estimates. The latter could be interpreted as evidence against the natural rate hypothesis, which has major economic policy implications. Due to its implications, further research should be encouraged in various avenues. First, estimates of potential GDP that take explicitly into account the role of hysteresis (if any) should be developed. Second, hysteresis effects are likely to be asymmetric, thus, non-linear effects should also be studied. Finally, if hysteresis is indeed found and becomes a well-established fact, a major revision to economic policy should be done.

References

Ball, L., & Onken, J. (2022). Hysteresis in unemployment: Evidence from OECD estimates of the natural rate. International Finance, 25 (3), 268-284. Retrieved from https://onlinelibrary.wiley.com/doi/abs/10.1111/infi.12424 doi: https://doi.org/10.1111/infi.12424

Blanchard, O. (2018, February). Should we reject the natural rate hypothesis? Journal of Economic Perspectives, 32 (1), 97-120. Retrieved from https://www.aeaweb.org/articles?id=10.1257/jep.32.1.97 doi: 10.1257/jep.32.1.97

Cerra, V., Fatás, A., & Saxena, S. C. (2023, March). Hysteresis and business cycles. Journal of Economic Literature, 61 (1), 181-225. Retrieved from https://www.aeaweb.org/articles?id=10.1257/jel.20211584 doi: 10.1257/jel.20211584

DeLong, J. B., Summers, L. H., Feldstein, M., & Ramey, V. A. (2012). Fiscal policy in a depressed economy [with comments and discussion]. Brookings Papers on Economic Activity, 233-297.

Friedman, M. (1968). The role of monetary policy. The American Economic Review , 58 (1), 1–17. Retrieved 2022-04-04, from http://www.jstor.org/stable/1831652

Jordà, O , Singh, S. R., & Taylor, A. M. (2020, January). The long-run effects of monetary policy (Working Paper No. 26666). National Bureau of Economic Research. Retrieved from http://www.nber.org/papers/w26666 doi: 10.3386/w26666

Rawdanowicz, L, Bouis, R., Inaba, K.-I., & Christensen, A. K. (2014). Secular stagnation: Evidence and implications for economic policy. (1169). Retrieved from https://www.oecd-ilibrary.org/content/paper/5jxvgg6q27vd-en doi: https://doi.org/https://doi.org/ 10.1787/5jxvgg6q27vd-en

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